

MCUs And PSoCs In ELECTRONIC TOY APPLICATIONS



Ronak Desai
is system
engineering
manager
at Cypress
Semiconductor,
Bengaluru

Microcontrollers (MCUs) in current electronic toy applications are mainly used for infrared (IR) interfaces, external EEPROM interfaces, analogue inputs using external analogue-to-digital converters (ADCs) and comparators, timer and clock interfaces, monitoring battery inputs, controlling front panel mechanical keypads and displaying the results on LCDs/LEDs/graphical displays.

Programmable systems on chip (PSoC) devices can be used in remote control systems as single-board solutions for IR interfaces, battery input detection, analogue measurements and direct-drive LCD/graphical displays. Such devices support CapSense technology, which replaces mechanical buttons on the keypad.

Using IDE tool of SoC devices, all interfaces and logics can be designed. IDE tools

have a readily-available component block for designing the interface and logic like IR, ADC for analogue inputs, pulse-width modulator (PWM) for buzzer applications, DACs, segments, characters or graphical LCD display support.

This article discusses the role of MCUs and PSoCs in electronic toy applications. It also evaluates system limitations and design challenges for these applications. With PSoCs, devices can reduce the complete product cost (by reducing bills of material cost) and project cost (by implementation in IDE tool) in electronic toy applications.

Theory behind

Current electronic toy applications use 8-bit/16-bit MCUs for different functions like motor control, LED/LCD/graphical display drive and different input and output interfaces, battery sensing and audio output. The MCU in CSP package is most commonly used in electronic toy applications.

The MCU receives analogue input signals from analogue sensors (sensors that receive analogue inputs and provide digital voltage that is applied to the MCU) and 2-axis/3-axis accelerometer (accelerometer measures the 2-axis/3-axis movement and converts it to digital voltage, which is applied to the MCU). The MCU also uses the accelerometer and mechanical joystick for running gaming applications.

The MCU uses external buffer driver circuitry to drive LED, LCD or graphical displays. Typically, an LCD or a graphical display with backlight is used in low-end applications. The MCU also controls the backlight of the LCD and graphical display. It uses a touchscreen controller for touchscreen interface in high-end applications.

The MCU uses a mechanical keypad for user inputs in low-end applications. It uses CapSense/touchscreen based keypads in high-end applications. The MCU interfaces

Fig. 1: Block diagram using an MCU

